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Technical proposal (Envelope 1)

Tender NGPA/T/2026/031 · Smart Gate Automation System for the Terminal 2 Truck Gates

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/031, we, Sahel Gate Technologies L.L.C., submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Mariam Al-Kuwari
General Manager, Sahel Gate Technologies L.L.C.

Date: 19 October 2026
Reference: SGT-T2G-2026-11

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1. Company and references

Sahel Gate Technologies is a systems integrator founded in 2009 in Al-Noor, with 85 staff. We specialise in terminal gates, access control and CCTV for ports and free zones. We are the certified partner in the State for the ClearLane OCR product line, and we run a service centre and spare-parts store in the Al-Noor Logistics Park, 6 km from Terminal 2.

Reference	Scope	Year
Port of Ras Hadd, Container Terminal	8 lanes: OCR, RFID, Gate Operating System, TOS integration	2022
Free Zone Logistics Gate	4 lanes with kiosks, Vehicle Booking System integration	2024
Al-Waha Inland Container Depot	3 lanes: OCR and damage imaging	2025

2. Lane equipment

Each lane gets a ClearLane OCR portal: a galvanised steel gantry with 2 plate cameras (front and rear), 4 container-number cameras and 5 damage-inspection cameras of 12 megapixels covering the four sides and the roof, with infrared illumination for night operation. Cameras are rated IP67 and the lane cabinets IP66, for operation from 0 °C to +55 °C, with marine-grade coating for the coastal atmosphere.

Measured OCR performance. Over 12 months at the Port of Ras Hadd (1.9 million transactions), container numbers were read correctly in 99.1 % of transactions, with ISO 6346 check-digit validation, and truck plates with Arabic and Latin characters in 98.2 %. Readings are valid at speeds up to 20 km/h, by day and night.

Damage images. Images are stored on the gate storage array and kept online for 60 days. Older images are moved to the Authority's backup system and can be restored on request.

RFID. Two UHF readers per lane (ISO/IEC 18000-63) and 3,000 tamper-evident windscreen tags that break when removed.

Kiosks. Stainless-steel kiosk rated IP66, with a 15-inch sunlight-readable touchscreen, Arabic and English interface, an intercom to the gate control room and a thermal printer for the gate pass with the assigned yard position.

Transaction time. At Ras Hadd, the average in-gate time of a pre-booked truck without exceptions is 38 seconds from arrival at the portal to the barrier opening.

3. Gate Operating System and integrations

The ClearLane Gate Operating System controls the lanes, barriers, traffic lights and kiosks. Exceptions (no appointment, unreadable number, damage, no customs release) go to the exception workstation, which shows all images and readings of the transaction. Dashboards show transactions per hour, average transaction time per lane and exceptions by type.

The GOS checks every visit against the appointment in the PCS Vehicle Booking System (REST API) and against the customs release status received through the PCS. The TOS integration uses REST web services and UN/EDIFACT CODECO. The barrier opens only when the appointment, the TOS work order and, for exits, the customs release are valid.

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4. Architecture and security

Two GOS application servers in a cluster and a storage array in the Terminal 2 server room, and a standby server in the Authority's main data centre with database replication. Two UPS units with 60 minutes of autonomy and redundant network switches per lane group. We commit to 99.7 % availability per calendar month. If the GOS is unavailable, the kiosks switch to manual mode and the control room opens the barriers, with the OCR readings still displayed.

- All equipment sits on a dedicated OT network segment behind the Authority's firewalls, with no internet access. All data stays in the Authority's server rooms; no cloud service is used.
- Users sign in with their Active Directory accounts; administrator accounts use multi-factor authentication.
- Audit logs of all user and system events are kept for 3 years and forwarded in real time to the Authority's SIEM in syslog format.
- Interfaces use REST/JSON and UN/EDIFACT. The API documentation and the GOS database schema are delivered to the Authority.

5. Implementation, training and maintenance

Milestone	Month
Design approval	1
Civil works and portal foundations	2–3
Lanes 1 and 2 in operation	4
Lanes 3 and 4 in operation	5
Lanes 5 and 6 in operation, Site Acceptance Test	6
30-day performance test and acceptance	7

At least 4 lanes stay open throughout the works.

Training. 20 gate clerks and control-room operators (3 courses) and 6 maintenance technicians (2 weeks, including on-the-job training). Courses are given in Arabic and English. The operator manual is provided in Arabic and English; the maintenance and technical manuals are in English.

Warranty and maintenance. 24-month warranty from acceptance, parts and labour. Three years of maintenance after the warranty: quarterly preventive maintenance, software updates, a 24/7 hotline, and on-site response within 4 hours from our Al-Noor service centre, which holds a stock of critical spare parts (cameras, readers, barrier controllers, kiosk printers).

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6. Compliance matrix (Annex 1)

Clause	Requirement	Response	Offered
3.1	OCR accuracy: containers ≥ 98 %, plates (Arabic and Latin) ≥ 97 %, up to 15 km/h	Comply	99.1 % containers, 98.2 % plates, up to 20 km/h (§2)
3.2	Damage images, 4 sides and roof, ≥ 5 MP, 90 days online	Comply	5 cameras of 12 MP; online storage and archive (§2)
3.3	UHF RFID ISO/IEC 18000-63, 3,000 tags	Comply	2 readers per lane, 3,000 tags (§2)
3.4	Average in-gate transaction ≤ 60 s	Comply	38 s average at Ras Hadd (§2)
3.5	Kiosk: Arabic and English, intercom, gate pass	Comply	§2
3.6	GOS: lane control, exceptions, dashboards	Comply	§3
3.7	Integration: TOS, PCS Vehicle Booking System, customs release	Comply	TOS, VBS and customs release (§3)
3.8	On-premise in T2 server room and main data centre; POTS-2026	Comply	T2 server room + main data centre (§4)
3.9	Availability ≥ 99.5 %; manual mode	Comply	99.7 %, manual mode (§4)
3.10	IP66, 0 °C to +55 °C	Comply	IP66/IP67, 0 to +55 °C (§2)
4.1	All 6 lanes in operation within 8 months	Comply	Month 7 (§5)
4.2	SAT and 30-day performance test	Comply	§5
4.3	Training: 20 operators, 6 technicians	Comply	§5
5.1	Warranty 24 months	Comply	24 months (§5)
5.2	3 years of maintenance after warranty	Comply	§5
5.3	On-site within 4 h, 24/7; spare parts in the State	Comply	Al-Noor service centre (§5)

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